ARCHITECTURAL PARADIGM / FORMAL SPECIFICATION

JTK

A deterministic architecture of reciprocal feedback

AC excitation. Closed causal loops. Self-reference.
Internal reconstruction. Exact finite LUT execution.

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DEDICATED TO

Jitske Koenders

For Jitske, to whom this architecture is dedicated.

$$\mathcal{T}_K(s_k, i_k) = (s_{k+1}, o_k)$$

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The architectural brief supplies the organizing ideas. This edition supplies explicit operators, domains, update order, analog equations and a complete finite reference realization.

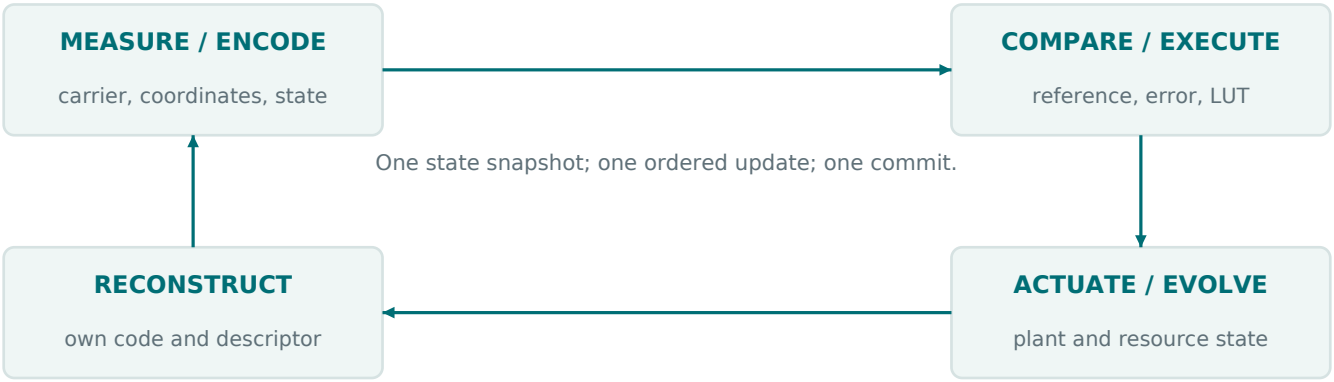
The closed architectural loop

A single state transition connects observation, action and the reconstruction of its own organization.

JTK is a parameterized architecture in which a process is measured, encoded, compared with a reference, acted upon and represented internally. That representation participates in the next transition. The rulebook **K** fixes every type, operator, table and update order.

$$\mathcal{T}_K : \mathcal{S} \times \mathcal{I} \longrightarrow \mathcal{S} \times \mathcal{O} \tag{1}$$

$$s_k = (x_k, z_k, c_k, d_k, p_k, \boldsymbol{\tau}_k, \boldsymbol{\mu}_k, E_k) \tag{2}$$



Input and resource ports enter the declared state transition.

State component	Meaning
x, z	Process state and controller memory.
c, d	Executable organization and its constructive description.
p, τ, μ	Carrier phase, internal process phases and their increments.
E; i; o	Resource account; accepted input record; emitted output record.

Closure means that every accepted state/input pair produces exactly one successor in the same declared state space. **Causality** means that the measured state at tick k determines an action whose effects enter the successor state. Snapshot-and-commit execution follows the synchronous approach [5].

For a driven instance, the initial state and complete input trace determine the trajectory. An autonomous instance includes its input generator in the state. Coupling is expressed by the declared dependency graph and its ports.

Self-reference and reconstruction

The system carries a constructive description of the organization that executes its transition.

Choose finite description and organization sets D and C, with a constructor B_K: D → C and an exact encoding satisfying $Enc_K(B_K(d)) = d$. The rulebook K supplies the interpretation of these descriptions. Initialize $c_0 = B_K(d_0)$.

$$y_k = H_K(x_k, c_k, p_k, i_k), \qquad v_k = E_K(y_k, d_k, \boldsymbol{\tau}_k) \tag{3}$$

$$u_k = L_{c_k}[A_K(v_k, z_k)], \qquad x_{k+1} = P_K(x_k, u_k, i_k) \tag{4}$$

$$z_{k+1} = Z_K(z_k, y_k, u_k, i_k) \tag{5}$$

$$d_{k+1} = D_K(d_k, c_k, x_k, z_k, y_k, u_k, E_k, i_k) \tag{6}$$

$$c_{k+1} = B_K(d_{k+1}) \tag{7}$$

These are ordered evaluations on the old-state snapshot, followed by one commit. The constructor consumes the newly calculated description. Tables, geometric boundaries and routing coefficients can all be components of c.

Operational autopoiesis

Here, operational autopoiesis is the recurring internal production of the executable organization from its constructive description and resource state. The loop is description → construction → execution → revised description. This is the engineering definition used in this specification; self-production supplies its conceptual foundation [4].

The paired J operator

Let S exchange two channels: S(a,b) = (b,a). The shared-half operator J and its complementary component are:

$$J = \frac{I+S}{2}, \quad J(a,b) = \left(\frac{a+b}{2}, \frac{a+b}{2}\right), \quad J^2 = J \tag{8}$$

$$v = Jv + (I-J)v, \qquad J(I-J) = 0 \tag{9}$$

Both components together retain the pair exactly. Rational numerators or a retained fractional bit represent the half operation exactly. J names the paired decomposition; T names the complete causal transition.

OTAN2: reversible log coordinates

Canonical profile SLOG2: the componentwise signed-log angle from the architectural brief, completed by its compatible radius and inverse.

Let $\ell > 0$ be a reference length. For Cartesian coordinates x, y, z, choose a vertical origin z_0 and height $H_z > 0$. Each logarithm has a dimensionless argument.

$$g_\ell(a) = \operatorname{sgn}(a)\ln\left(1 + \frac{|a|}{\ell}\right), \qquad g_\ell(0) = 0 \tag{10}$$

$$u = g_\ell(x), \quad v = g_\ell(y), \quad R_w = \sqrt{u^2 + v^2} \tag{11}$$

$$\theta_w = \operatorname{OTAN2}(y, x) = \operatorname{atan2}(v, u) \bmod 2\pi \tag{12}$$

At x = y = 0, define $\theta_w = 0$. Elsewhere θ_w lies in [0, 2π). The complete nonlinear coordinate map is:

$$\mathcal{W}(x, y, z) = (R_w, \theta_w, \zeta), \qquad \zeta = \frac{z - z_0}{H_z} \tag{13}$$

Exact inverse

$$u = R_w \cos\theta_w, \qquad v = R_w \sin\theta_w \tag{14}$$

$$x = \ell \operatorname{sgn}(u)(e^{|u|} - 1), \quad y = \ell \operatorname{sgn}(v)(e^{|v|} - 1) \tag{15}$$

$$z = z_0 + H_z \zeta \tag{16}$$

The image uses the single canonical angle 0 at $R_w = 0$. With this convention, W and its inverse recover each Cartesian point. For example, (x,y) = (ℓ,0) maps to (R_w,θ_w) = (ln 2,0), and the inverse returns (ℓ,0).

Optional radial profile LOGR

$$\rho = \sqrt{x^2 + y^2}, \quad r_{\log} = \ln(1 + \rho/\ell), \quad \theta = \operatorname{atan2}(y, x) \bmod 2\pi \tag{17}$$

LOGR retains the ordinary geometric angle; SLOG2 uses the warped component angle. The profile identifier fixes which geometry is meant. Forward encoding is logarithmic and the inverse map is exponential.

PHI, Fibonacci cuts and CSG

Physical radius is selected before OTAN2: √(x²+y²) for planar/conical cuts, or √(x²+y²+z²) for spherical cuts.

$$\Phi = \frac{1+\sqrt{5}}{2}, \qquad F_1 = F_2 = 1, \quad F_{n+2} = F_{n+1} + F_n \tag{18}$$

Finite Fibonacci shells

For radius R > 0 and integer N ≥ 1, define increasing boundaries $r_0 = 0$ and:

$$r_i = R\frac{F_{i+1}}{F_{N+1}}, \qquad i = 1, \ldots, N \tag{19}$$

Use $[r_{i-1}, r_i)$ for each shell and include R in the final shell. Rational R gives rational endpoints. For N = 6, the radii are R times 0, 1/13, 2/13, 3/13, 5/13, 8/13 and 1.

Golden shells and the golden spiral

$$\rho_j = R\Phi^{-j} \ (0 \le j \le M), \quad \ln\rho_j - \ln\rho_{j+1} = \ln\Phi \ (j < M) \tag{20}$$

$$\rho(\theta) = a\exp\left(\frac{2\ln\Phi}{\pi}\theta\right), \quad \rho(\theta + \pi/2) = \Phi\rho(\theta) \tag{21}$$

Here M ≥ 1 is an integer and a > 0. Golden shells use $(\rho_{j+1}, \rho_j]$ for j = 0,...,M-1, plus the core $[0, \rho_M]$. Fibonacci shells, golden shells and the spiral are named profiles; intersect their cells with the selected solid.

Hemisphere, cone and set difference

$$H_R = \{(x,y,z) : x^2 + y^2 + z^2 \le R^2, \ z \ge 0\} \tag{22}$$

$$C_{R,H} = \{(x,y,z) : 0 \le z \le H, \ H^2(x^2+y^2) \le R^2(H-z)^2\} \tag{23}$$

$$\chi_{A\cap B} = \chi_A\chi_B, \quad \chi_{A\cup B} = \max(\chi_A, \chi_B), \quad \chi_{A\setminus B} = \chi_A(1-\chi_B) \tag{24}$$

For these solids R,H > 0. Primitive equality boundaries belong to the primitive; B owns the removed boundary in A \ B. The cone apex (0,0,H) has planar radius zero and canonical angle. Rational coordinates reduce the solid predicates to exact integer comparisons after denominators are cleared.

Double Hadamard partitioning

A weighted tensor comparison selects one branch at every node, including every equality case.

At node n, use equally shaped real tensors X, $W_{1,n}$ and $W_{2,n}$, and a real threshold β_n. Let d be the number of tensor entries.

$$\sigma_n(X) = \sum_{i,j} (W_{1,n})_{ij} X_{ij} (W_{2,n})_{ij} - \beta_n \tag{25}$$

$$\sigma_n(X) = \langle W_{1,n} \odot W_{2,n}, X \rangle_F - \beta_n \tag{26}$$

Score	Branch	Stored consequence
$\sigma < 0$	Left	Continue to the specified left child or leaf.
$\sigma \geq 0$	Right	Continue to the specified right child or leaf.

The double Hadamard product forms elementwise weighted features. Summation uses all selected entries. A leaf may hold a CSG occupancy label, a kinematic action index or a channel assignment. The labels “lipid” and “carbohydrate” may identify boundary and resource roles in a selected adapter.

Integer arithmetic with a sufficient accumulator

For integer tensors with bounds $|X_{ij}| \leq X_{\max}$, $|W_{1,ij}| \leq A$, $|W_{2,ij}| \leq B$ and $|\beta| \leq B_0$:

$$|\sigma| \leq M = dABX_{\max} + B_0 \tag{27}$$

$$w_{\mathrm{acc}} = 1 + \lceil \log_2 (M+1) \rceil \tag{28}$$

A signed accumulator of this width contains every score exactly. Specify integer products before any rescaling. Rational weights can share a common denominator and be compared by their integer numerators.

Execution cost

$$\text{work} = O(dh), \qquad h = O(\log N_{\text{leaf}}) \text{ for a balanced tree} \tag{29}$$

Fix a maximum depth h in the profile. A schedule requiring equal work per request can pad shorter paths to h stages while retaining the selected leaf. The geometry, leaf payload and schedule are compiled components of c.

Boundary convention. Zero goes right. Shell intervals are half-open except the terminal interval. Primitive equality belongs to the primitive. These conventions make the composed spatial encoder single-valued.

Packed geometry and complete LUTs

Quantization, addressing and byte interpretation are part of the mathematical specification.

For SLOG2, choose $R_{w,max} > 0$ and use the coordinate domain $0 \le R_w \le R_{w,max}$, $0 \le \theta_w < 2\pi$, $0 \le \zeta \le 1$. The source layout has 12 angular bits, 12 radial bits and 8 vertical bits.

$$q_\theta = \left\lfloor \frac{4096\theta_w}{2\pi} \right\rfloor, \quad q_R = \min\left(4095, \left\lfloor \frac{4096 R_w}{R_{w,\max}} \right\rfloor\right) \tag{30}$$

$$q_z = \min(255, \lfloor 256\zeta \rfloor), \qquad A = 2^{20} q_\theta + 2^8 q_R + q_z \tag{31}$$

Field	Bits	Mask
θ_w bin	31..20	0xFFF00000
R_w bin	19..8	0x000FFF00
ζ bin	7..0	0x000000FF

For each finite input code v, define E[v] = Q(W(Decode(v))). The stored integer E[v] is the runtime encoder result. Q defines the bins; the original payload can travel beside its address when exact recovery of the input record is required.

$$2^{32} \text{ entries} \times 8 \text{ bytes} = 2^{35} \text{ bytes} = 32 \text{ GiB} \tag{32}$$

A complete transition table

Enumerate finite states and inputs by bijections i_S and i_I. Store each next state and output at one address:

$$A_T(s,i) = |\mathcal{I}|\, i_S(s) + i_I(i), \qquad L_T[A_T(s,i)] = \mathcal{T}_K(s,i) \tag{33}$$

A full binary table with s state bits, a input bits and o output bits occupies $2^{s+a}(s + o)$ bits. Factorized tables implement the same transition by composition and can use less storage.

Canonical execution rules

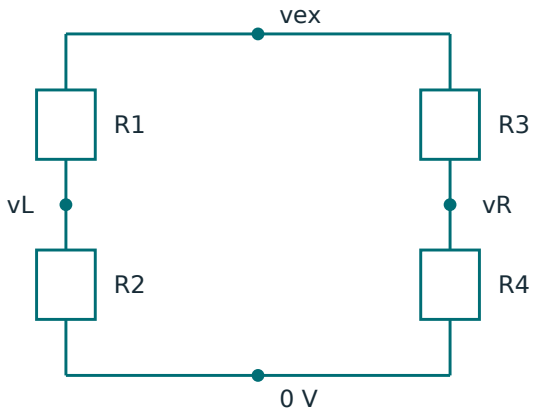
State fields have declared widths and signedness. Addition and multiplication use sufficient intermediates; division uses mathematical floor unless a rational pair is retained. Only named phase counters wrap. Named bounded fields saturate. Records use little-endian byte order, fixed field order and one committed update per tick.

Exactness means equality to the declared discrete function and its stored entries. A geometry LUT and a full transition LUT are separate tables with separate address spaces. The complete finite profile on pages 13-17 supplies actual table bytes.

The strain-gauge bridge

AC denotes alternating-current excitation. The feedback controller closes the measured process loop.

Define strain ε = ΔL/L_0 and gauge factor g by ΔR/R = gε. Use four positive resistances and a readout of infinite input impedance. The numbering below fixes the output sign [1,2].



Output polarity: left midpoint minus right midpoint.

$$b = \frac{R_2}{R_1 + R_2} - \frac{R_4}{R_3 + R_4}, \qquad v_{\mathrm{out}}(t) = b\, v_{\mathrm{ex}}(t) \tag{34}$$

The coefficient b remains defined when the AC excitation crosses zero. Balance occurs when $R_2R_3 = R_1R_4$.

Quarter bridge: exact constitutive relation

Set $R_2 = R(1 + g\varepsilon)$ and $R_1 = R_3 = R_4 = R$, with R > 0, g ≠ 0 and gε > -1.

$$b = \frac{g\varepsilon}{4 + 2g\varepsilon}, \qquad \varepsilon = \frac{4b}{g(1 - 2b)}, \qquad -\frac{1}{2} < b < \frac{1}{2} \tag{35}$$

Complementary full bridge

$$R_1 = R_4 = R(1 - g\varepsilon), \qquad R_2 = R_3 = R(1 + g\varepsilon) \tag{36}$$

$$b = g\varepsilon, \qquad |g\varepsilon| < 1 \tag{37}$$

These identities follow directly from the defined resistor model. A physical adapter records its calibrated gauge factor, excitation, gain, temperature dependence and lead contribution. Those parameters determine the observation function H_K.

Carrier and synchronous recovery

The excitation and demodulation reference share a defined phase and integration window [2,3].

$$v_{\mathrm{ex}}(t) = V_c \cos(\omega_c t), \qquad V_c > 0, \quad \omega_c > 0 \tag{38}$$

$$v_s(t) = G V_c b \cos(\omega_c t + \phi) + v_o + n(t), \qquad G > 0 \tag{39}$$

For constant b and constant offset v_o over M complete carrier periods, let $T = 2\pi M/\omega_c$, $M \geq 1$. Define the recovered bridge coefficient by:

$$\hat{b} = \frac{2}{G V_c T} \int_{t_0}^{t_0+T} v_s(t) \cos(\omega_c t)\, dt \tag{40}$$

$$\hat{b} = b \cos\phi + \eta_c, \qquad \eta_c = \frac{2}{G V_c T} \int_{t_0}^{t_0+T} n(t) \cos(\omega_c t)\, dt \tag{41}$$

The constant offset integrates to zero. Phase alignment $\varphi = 0$ and zero carrier projection $n_c = 0$ give $\hat{b} = b$. For changing strain, the same integral defines a windowed observation. The input record can retain phase and projected residuals explicitly.

Exact signed-carrier implementation

Choose an even N and a balanced carrier $a_j \in \{-1,+1\}$ with $\Sigma a_j = 0$. For recorded integer samples d_j, calculate:

$$D = \sum_{j=0}^{N-1} a_j d_j, \qquad \bar{b} = \frac{D}{NA} \tag{42}$$

$$d_j = A a_j b + o + n_j \quad \Longrightarrow \quad \bar{b} = b + \frac{1}{NA} \sum_{j=0}^{N-1} a_j n_j \tag{43}$$

For integer scale $A \neq 0$, store the result as numerator D and denominator NA, or apply the profile's declared integer rounding rule. A b-bit signed sample accumulator for N terms needs $b + \lceil \log_2 N \rceil + 1$ bits to include sign reversal of the most-negative sample.

Bridge-to-strain propagation

$$|\hat{\varepsilon} - \varepsilon| \leq \frac{4}{|g|(1 - 2b_{\max})^2} |\hat{b} - b|, \qquad b_{\max} < \frac{1}{2} \tag{44}$$

This bound applies when b and $\hat{b}$ both lie in $[-b_{max}, b_{max}]$. It follows from the exact quarter-bridge inverse. Measurement residual and computational representation therefore have separate, explicit quantities.

A force-balance realization

A mechanical state, a bridge observation and a controller form a complete analog feedback model.

Use a mechanical displacement q, mass m > 0, damping c > 0, stiffness k and actuator gain k_a. The calibrated instantaneous observation is $y = k_s q$. The external force is a declared input.

$$m\ddot{q} + c\dot{q} + kq = f_{\text{ext}} + k_a u, \qquad y = k_s q \tag{45}$$

$$e = r - y, \qquad \dot{z} = e, \qquad u = k_p e + k_i z \tag{46}$$

For this continuous PI profile, the closed-loop characteristic polynomial is:

$$ms^3 + cs^2 + (k + k_a k_s k_p)s + k_a k_s k_i \tag{47}$$

The cubic is Hurwitz precisely when the following inequalities hold, given m > 0 and c > 0:

$$k + k_a k_s k_p > 0, \quad k_a k_s k_i > 0, \quad c(k + k_a k_s k_p) > m k_a k_s k_i \tag{48}$$

For example, normalized coefficients $m = c = k = k_a = k_s = k_p = k_i = 1$ give $s^3 + s^2 + 2s + 1$ and satisfy $2 > 1$. This is a concrete stable continuous profile.

Null balance and dynamic force

For r = 0, nonzero k_a and k_s, and an attainable stable equilibrium with constant external force:

$$q_\infty = 0, \qquad u_\infty = -f_{\text{ext}}/k_a \tag{49}$$

$$f_{\text{ext}} = m\ddot{q} + c\dot{q} + kq - k_a u \tag{50}$$

The balancing action is the static force readout. The second relation retains inertia and damping for dynamic inference. An implementation including a demodulator or filter gives those devices their own states in x.

Deterministic continuous semantics

$$\dot{x} = f_K(x, u, t), \qquad x(t_0) = x_0 \tag{51}$$

Specify the initial state, coefficients and complete forcing. With a continuous, locally state-Lipschitz vector field and a compact forward-invariant state domain contained in the field’s domain throughout the operating interval, the flow is unique and continues over that interval [6]. This flow defines the analog model.

Master time and internal clocks

The master tick orders execution; internal phases encode the progress of the represented processes.

$$t_k = t_0 + k\Delta, \quad \Delta > 0, \qquad p_{k+1} = (p_k + 1) \bmod N_c \tag{52}$$

Choose positive integer moduli N_c, N_B and N_E. An atomic reference can supply the physical timing adapter; integer k defines logical ordering. The paired clocks make τ, τ, μ, μ explicit: structural phase τ_B, resource phase τ_E and nonnegative increments μ_B, μ_E.

$$\tau_{a,k+1} = (\tau_{a,k} + \mu_{a,k}) \bmod N_a, \qquad a \in \{B, E\} \tag{53}$$

$$\mu_{a,k+1} = M_{a,K}(x_k, c_k, E_k, i_k), \qquad 0 \le \mu_{a,k+1} < N_a \tag{54}$$

Unwrapped internal progress is the sum of the increments; τ stores its cyclic phase. A continuous process-clock profile uses $d\tau_a/dt = \mu_a(x,c,E,t) \geq 0$. These are process-rate coordinates with declared units.

Analog-to-digital ordering

$$d_k = Q(h(x(t_k^-))), \qquad (s_{k+1}, o_k) = \mathcal{T}_K(s_k, d_k) \tag{55}$$

$$u(t) = U(o_k), \qquad t_k \le t < t_{k+1} \tag{56}$$

At each tick: capture, encode, calculate, commit and hold. All coincident events follow that order. Continuous evolution uses the committed command; the next capture observes its result. An exact linear held-input flow is:

$$x_{k+1} = e^{A\Delta} x_k + \int_0^{\Delta} e^{A(\Delta - s)} B u_k \, ds \tag{57}$$

Resource accounting

$$\dot{E} = P_{\text{in}} - P_{\text{sense}} - P_{\text{act}} - P_{\text{build}} - P_{\text{diss}} \tag{58}$$

For a physical energy account, E is in joules and every P is in watts. For an integer profile, E is a declared resource count and the update uses exact integer receipts and expenditures. Construction is one of the accounted processes. The environment supplies resources through named ports; the internal organization closes the production loop.

One architecture, typed adapters

A domain enters through its state, observation, action, unit system and transition law.

Domain	State and observation	Action and internal production
Digital computing	Registers, memory, phase and table codes.	State update; reconstruct tables or representations.
Analog circuits	Capacitor voltages, inductor currents, bridge output.	Drive currents or voltages; update calibration states.
Mechanics / robotics	Position, momentum, strain and geometric occupancy.	Force or torque; update trajectories and constraints.
Chemical model	Concentrations, membrane variables and substrate counts.	Reaction fluxes; boundary and catalyst production.
Dataflow systems	Typed records, version, queue depth and logical time.	Transform or route records; regenerate derived state.

Exact adapter condition

Let E_D encode domain state, I_D encode its input and F_D be one domain step. Exact correspondence is the commuting relation:

$$E_D(F_D(x,v)) = \pi_S \mathcal{T}_K(E_D(x), I_D(v)) \tag{59}$$

A finite abstraction satisfies this relation when every represented state/input cell has one successor code and T_K stores that code. Refine the state or retain additional input information to obtain that property. Every variable has an explicit type, port and causal role.

Energy-consistent physical coupling [7]

$$\dot{x} = (J_H - R_H)\nabla\mathcal{H} + G_H u, \qquad y = G_H^T \nabla\mathcal{H} \tag{60}$$

$$J_H^T = -J_H, \quad R_H = R_H^T \geq 0, \quad \dot{\mathcal{H}} = -(\nabla\mathcal{H})^T R_H \nabla\mathcal{H} + u^T y \tag{61}$$

Stored energy is $\mathcal{H}(x)$. Skew coupling J_H transfers energy, R_H dissipates it, and $u^T y$ is supplied power. J_H is distinct from the paired J projection.

Chemical adapter profile

$$\dot{\xi} = N v(\xi) + j_{\text{in}} - j_{\text{out}}, \qquad v_j(\xi) = k_j \prod_i \xi_i^{\nu_{ij}} \tag{62}$$

The profile supplies stoichiometry N, rate constants k_j, exponents ν_{ij} and fluxes. “Lipid” denotes the chosen boundary variables; “carbohydrate” denotes the chosen substrate/resource variables. Their coupling is specified by this reaction network and its observation map.

A complete reference realization

REF-480 is a fully enumerated, dimensionless projection of the architecture: carrier, controller, process clock, resource and self-representation.

$$s=(x,p,\tau,E,q)\in\{-2,-1,0,1,2\}\times\mathbb{Z}_4\times\mathbb{Z}_4\times\{0,1,2\}\times\{0,1\} \tag{63}$$

$$w=(r,d,b,f)\in\{-1,0,1\}^2\times\{0,1\}^2 \tag{64}$$

x is process state; p is carrier phase; τ is internal process phase; E is actuator resource; q is the controller's representation phase. Inputs are reference r, disturbance d, resource receipt b and coherent representation toggle f. Thus |S| = 480 and |I| = 36.

Measurement, LUT and self-reference

$$a(p)=(-1)^{\lfloor p/2\rfloor},\quad y_{\mathrm{raw}}=a(p)x,\quad \hat{x}=a(p)y_{\mathrm{raw}}=x,\quad e=r-\hat{x} \tag{65}$$

$$U=(-1,-1,-1,0,1,1,1),\qquad C_0=U,\quad C_1=-U \tag{66}$$

$$\hat{q}=q\oplus f,\qquad u_0=(-1)^{\hat{q}}C_{\hat{q}}[e+3]=U[e+3] \tag{67}$$

All arrays use zero-based indices. The stored representation selects a signed encoding and participates in its own decoding. Both encodings produce the same controller. Set u = u_0 when E > 0; set u = 0 when E = 0. This is the finite actuator resource law.

One simultaneous state commit

$$x^+=\min(2,\max(-2,x+u+d)),\qquad p^+=(p+1)\bmod 4 \tag{68}$$

$$\tau^+=(\tau+\mathbf{1}_{E>0})\bmod 4,\qquad E^+=\min(2,E+b-|u|) \tag{69}$$

$$q^+=0 \tag{70}$$

All right-hand sides use the old state. Receipt b becomes available in E^+. Since |u| ≤ E, the resource successor is nonnegative. Reconstructing $q^+ = 0$ returns the executable representation to its canonical signed bank.

Exact finite closure

Every one of the 17,280 state/input pairs has one defined successor. Every successor remains in S. The carrier and representation phases are explicit; signed multiplication, table indexing, resource accounting and canonical reconstruction use integers only.

Integer reference implementation

The following function and indices define the runtime relation. The complete generator is embedded in the PDF.

```
U = (-1, -1, -1, 0, 1, 1, 1)
C = (U, tuple(-v for v in U))

def step(s, w):
    x, p, tau, E, q = s
    r, d, b, f = w
    carrier = 1 if p < 2 else -1
    raw = carrier * x
    sensed = carrier * raw
    error = r - sensed
    observed_q = q ^ f
    sign = 1 if observed_q == 0 else -1
    decoded = sign * C[observed_q][error + 3]
    u = decoded if E > 0 else 0
    xn = min(2, max(-2, x + u + d))
    pn = (p + 1) % 4
    taun = (tau + int(E > 0)) % 4
    En = min(2, E + b - abs(u))
    qn = 0
    return ((xn, pn, taun, En, qn),
            (raw, sensed, error, u, observed_q))

def state_index(s):
    x, p, tau, E, q = s
    return ((((x+2)*4+p)*4+tau)*3+E)*2+q

def input_index(w):
    r, d, b, f = w
    return (((r+1)*3+(d+1))*2+b)*2+f

def address(s, w):
    return 36*state_index(s) + input_index(w)
```

States and inputs are enumerated in ascending lexicographic product order. State indices are 0..479; input indices are 0..35. The mixed-radix address covers 0..17279 exactly once.

$$L[A] = \mathrm{step}(s, w), \qquad A = 36\, i_S(s) + i_I(w) \tag{71}$$

The full table stores ((next state), (raw, sensed, error, action, representation phase)). The compact table stores only the next-state index as an unsigned 16-bit little-endian record. Its byte offset is 2A and its exact size is 34,560 bytes.

Twelve exact transitions

Initial state: (-2, 0, 0, 2, 0). Every value below is an integer produced by the reference implementation.

k	x	r	d	b	f	E	u	x+	E+	p+	τ+
0	-2	1	0	1	0	2	1	-1	2	1	1
1	-1	1	0	1	0	2	1	0	2	2	2
2	0	1	0	1	0	2	1	1	2	3	3
3	1	1	1	0	0	2	0	2	2	0	0
4	2	1	0	0	1	2	-1	1	1	1	1
5	1	1	0	0	0	1	0	1	1	2	2
6	1	-1	0	0	0	1	-1	0	0	3	3
7	0	-1	0	1	0	0	0	0	1	0	3
8	0	-1	0	1	0	1	-1	-1	1	1	0
9	-1	-1	-1	0	0	1	0	-2	1	2	1
10	-2	-1	0	0	0	1	1	-1	0	3	2
11	-1	-1	0	1	0	0	0	-1	1	0	2

Ticks 0-2: the controller takes x from -2 to the reference 1. Resource receipts maintain E = 2. **Tick 3:** disturbance d = 1 moves x to 2. **Tick 4:** f = 1 selects the other signed representation, and decoding still produces u = -1.

Ticks 6-8: the reference becomes -1. The first action consumes the remaining resource. The next tick receives one unit; the following tick uses it to complete the move. **Ticks 9-10:** a negative disturbance is followed by a positive restoring action.

Clock and representation sequence

The last two columns give committed carrier and process phases. The carrier advances every tick. The process phase advances exactly when old E > 0; it remains at 3 during tick 7 and at 2 during tick 11. Every committed q is 0, while the read phase at tick 4 is 1.

$$y_{\mathrm{raw}} = a(p)x, \qquad \hat{x} = x, \qquad \hat{q} = f \text{ along this trace} \tag{72}$$

One loop, two phase histories. The clocks share an update boundary while retaining different phase histories. Resource receipt, disturbance and representation changes are explicit inputs to the same transition relation.

Determinism and closure

The finite realization has direct proofs and an exhaustive evaluation over its complete declared domain.

1. One successor for every accepted pair

The error e = r - x lies in {-3,...,3}; therefore e + 3 always selects one of seven entries. Every remaining operation is integer addition, multiplication, XOR, comparison, minimum, maximum or positive-modulus reduction. Each is single-valued on the declared domains. Thus T_K is total and deterministic. Induction gives a unique trajectory from s_0 and the input trace.

2. The state space is invariant

The bounded update places x^+ in {-2,...,2}. The two modulo operations place p^+ and τ^+ in Z_4. The actuator law gives |u| ≤ E, so 0 ≤ E + b - |u|; the upper bound gives $E^+ \le 2$. Finally $q^+ = 0$. Therefore $T_K(S \times I)$ has its successor state in S.

3. Representation and carrier identities

$$a(p)^2 = 1, \qquad (-1)^{\hat{q}} C_{\hat{q}}[e+3] = U[e+3] \tag{73}$$

The sensed integer equals x, and both signed controller encodings give identical actions and successor dynamics. Canonical reconstruction is idempotent: applying q → 0 a second time returns the same q.

4. Finite-time regulation

Let r be constant, d = 0 and b = 1 each tick. At most one initial tick is needed to obtain E ≥ 1. Thereafter each nonzero action decreases |r - x| by exactly one. At x = r the control is zero, and the target remains invariant. Representation toggles preserve this result.

$$k_{\text{hit}} \le |r - x_0| + 1 \le 4 \tag{74}$$

5. Exact LUT equivalence

The mixed-radix indices are bijections. Defining each row as L[A(s,w)] = step(s,w) makes table execution equal to direct execution on every pair. Factored carrier, controller and normalization tables compose to the same relation.

Finite evaluation	Result
Complete state/input domain	480 × 36 = 17,280 transitions
Distinct complete-table addresses	17,280
Initial-state / constant-target cases	1,440; maximum first arrival: 4 ticks
Compact binary representation	34,560 bytes; exact uint16 LE index round trip

The evaluated object is REF-480 as defined on pages 13-14. Its complete transition relation, serialization and execution trace are embedded with this edition.

Configuration and table identity

The configuration records the exact constants and encoding of REF-480. Author and dedication fields are document metadata.

```
[identity]
author = Tom Klootwijk
identifier = NL200678942
date_of_birth = 10-07-1990
dedication = Jitske Koenders

[reference]
profile = JTK-REF-480-v1
state_order = x,p,tau,E,q
input_order = r,d,b,f
initial_state = -2,0,0,2,0
x_min = -2
x_max = 2
phase_modulus = 4
process_modulus = 4
resource_capacity = 2
canonical_representation = 0
controller_U = -1,-1,-1,0,1,1,1
bank_0 = U
bank_1 = -U

[execution]
arithmetic = integers
commit = simultaneous_after_ordered_evaluation
enumeration = ascending_lexicographic_product
address = 36*state_index+input_index
next_state_record = uint16_little_endian
next_state_records = 17280
next_state_bytes = 34560
```

SHA-256 of the authoritative tables

Compact next-state table (34,560 bytes):

```
d377e15a6bdfdfcf33e30ae198e8181c96b2ce2c84390b561bcdbc0ead9f3b59
```

Full canonical JSON table (480,769 bytes):

```
f84b3057df2f3950bd4fd71addd14015ae7cba2c20a67cb873baffcc598b5618
```

Embedded files: jtk_reference.py; jtk_reference.ini; canonical_lut.json; next_state_lut_u16le.bin; finite_evaluation.json; jtk_equations.txt. The full JSON uses compact ASCII JSON with tuples represented as arrays.

Foundations and provenance

The equations, conventions and finite profile are stated directly in this edition; the following sources establish their technical background.

[A] Architectural source

Tom Klootwijk. *Pasted text.txt*, supplied for this edition. Source vocabulary: OTAN2, double Hadamard partitioning, PHI/Fibonacci cuts, paired J/T, internal clocks, packed kinematic LUTs and autopoietic organization.

[1] Strain and bridge measurement

NI. *Measuring Strain with Strain Gages.* Gauge factor, Wheatstone bridges, excitation and calibration. Used on page 8.

[2] AC bridge circuitry

Jim Williams. Analog Devices, *AN-43: Bridge Circuits.* AC-driven bridge and synchronous-demodulator sections; strain-gauge appendix. Used on pages 8-9.

[3] Synchronous demodulation

Brian Harrington. Analog Devices, *Low-Power Synchronous Demodulator Design Considerations.* Carrier phase, phase-sensitive detection and bandwidth. Used on page 9.

[4] Autopoietic organization

F. G. Varela, H. R. Maturana and R. Uribe (1974). *Autopoiesis: The organization of living systems, its characterization and a model.* BioSystems 5(4), 187-196. doi:10.1016/0303-2647(74)90031-8. Conceptual source for page 3.

[5] Synchronous execution

A. Benveniste and G. Berry (1991). *The synchronous approach to reactive and real-time systems.* Proceedings of the IEEE 79(9), 1270-1282. doi:10.1109/5.97297. Logical reaction and execution model, page 2.

[6] Unique continuous flow

Tom Hutchcroft (2024). Caltech, *Math 2: Ordinary Differential Equations.* Chapter 3, existence, uniqueness and maximal solutions. Used on page 10.

[7] Multiphysics energy balance

Arjan van der Schaft (2024). *Port-Hamiltonian nonlinear systems.* arXiv:2412.19673. Energy balance and coupling through physical ports. Used on page 12.

Notation. Superscript + denotes the next state. ⊙ is the Hadamard product; ⊕ is XOR; χ is a set indicator; Z_N is the cyclic phase set {0,…,N-1}; π_S selects the state component. Symbol meanings are local to the explicitly named domain.

Edition constructions. The compatible OTAN2 inverse, exact finite partitions, paired J projection, constructive transition schema, adapter relation and REF-480 specification complete the supplied architectural brief. All numeric table identities refer to this edition.